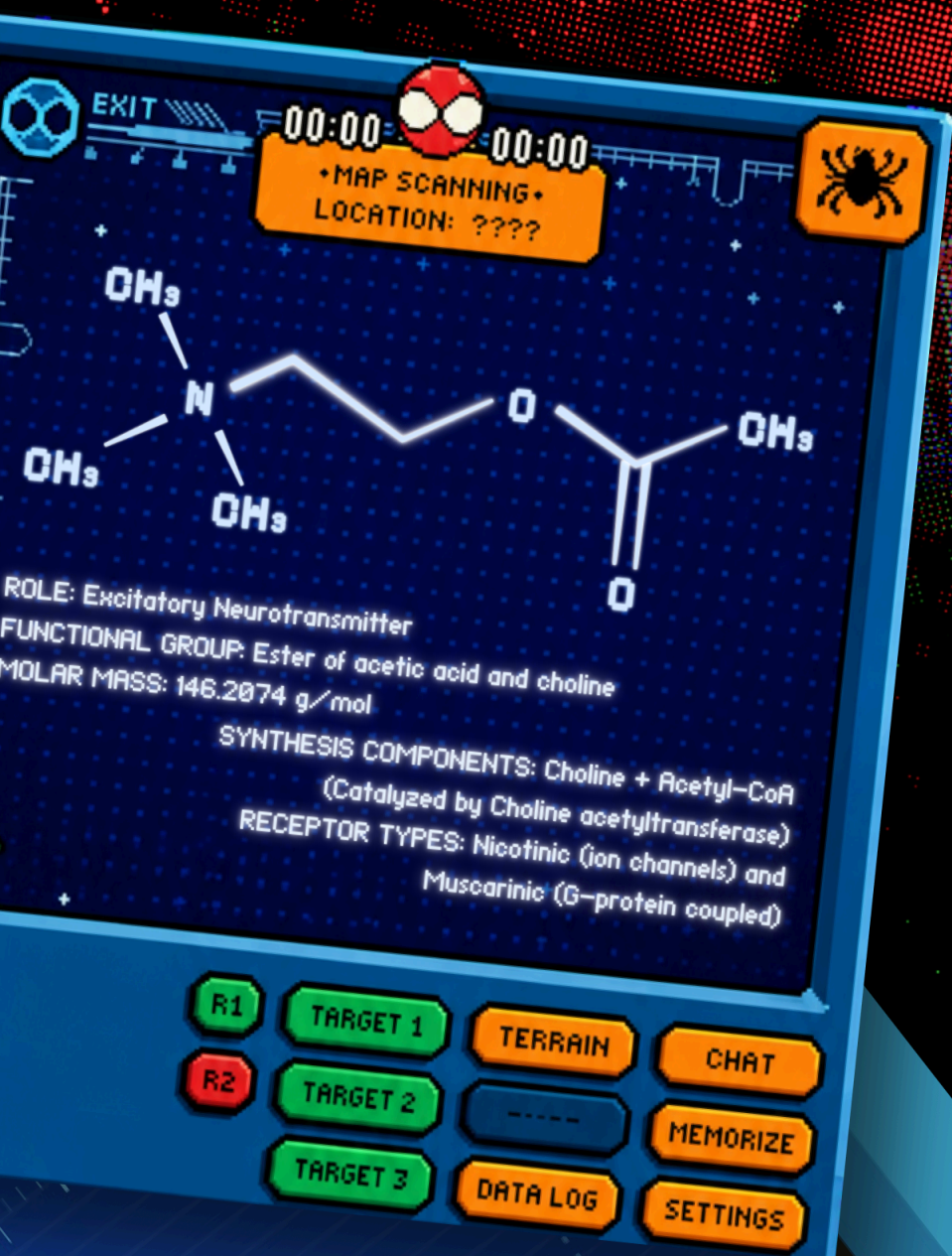


DIAZ, FRANCINE ANGEL B. ChE 2B

SPIDEY SENSES?



Ach

Acetylcholine ($C_7H_{16}NO_2$)

Spider-Man may sense danger before anyone else does—but behind every lightning-fast reaction, the real hero is not a radioactive spider bite, but a network of chemical signals called **neurotransmitters**. One of the best and oldest contenders is **Acetylcholine (ACh)**.

HOW & WHERE?

- At the **Neuromuscular Junction & Autonomic System**: ACh triggers skeletal muscle contractions for swift movement while regulating involuntary functions like heart rate, breathing, digestion, and secretions.
- Via **Nicotinic & Muscarinic Receptors**: It delivers rapid responses by opening ion channels (nicotinic) or slower cellular signals (muscarinic). Once transmitted, Acetylcholinesterase (AChE) instantly degrades ACh to stop continuous stimulation.

STEP-BY-STEP

- Stimulus
- Sensory receptor detects the stimulus, signal travels through a sensory neuron, and brain/spinal cord processes signal
- Signal travels through a motor neuron & reaches the neuromuscular junction
- ACh releases, binds to muscle receptors
- Muscle cell generates signal and contracts
- Physical Response



The Spidey tingle warns Spider-Man that danger is near. In our bodies, detecting danger triggers nerve signals that ultimately rely on acetylcholine (ACh) to tell muscles to respond.

But what happens when ACh itself becomes the problem?

Black widow spider venom can trigger the **massive release of ACh**, turning a messenger that normally enables controlled movement into a driver of excessive muscle stimulation.



Massive ACh release

Black widow venom can cause nerve endings to release unusually large amounts of acetylcholine.



Muscles become overstimulated

The sudden surge of ACh causes excessive stimulation of muscles, producing painful and sustained muscle contractions.



Neurotransmitter stores become depleted

With continued excessive release, available ACh can become depleted, disrupting normal communication between nerves and muscles.



Weakness and paralysis can follow

When normal nerve-to-muscle signaling can no longer be maintained, severe muscle weakness and, in serious cases, paralysis may occur.

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